KANGYOON LEE

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RESEARCH INTERESTS

Network Traffic Understanding and Control for AI Datacenters

My research aims to accelerate AI experimentation by reducing training job completion time on existing GPU clusters. I investigate how AI workload execution shapes inter-server GPU collective traffic. I seek to use this understanding to anticipate communication demand and design proactive routing and scheduling mechanisms that mitigate network bottlenecks.

EDUCATION

Aug. 2026 – Present — **Johns Hopkins University (JHU)** — Baltimore, MD
Ph.D. in Computer Science

Mar. 2018 – Feb. 2025 — **Hankuk University of Foreign Studies (HUFS)** — Yongin, Republic of Korea
B.Eng. in Information and Communications Engineering

- GPA: 4.34/4.50 (**Top 1.8%,** *Summa Cum Laude*)
- Thesis: *A Packet Scheduling Scheme Using Proportional-Fair Scheduling Algorithm in Edge Network*
- Completed 19 months of mandatory military service

RESEARCH EXPERIENCE

Aug. 2026 – Present — **Understanding MoE-Driven Collective Traffic in Distributed Training**
Graduate Research Assistant | Foundational Networked Systems Lab, JHU

- Investigating how MoE execution phases shape collective communication traffic by instrumenting DeepSpeed operations and aligning traces with NIC measurements
- Characterizing cross-rank collective entry/exit skew and traffic gaps through comparisons of MoE configurations and repeated training runs

Mar. 2025 – May 2026 — **Data Pipeline Development and Training Performance Profiling**
Full-time Research Intern | Korea Institute of Science and Technology (KIST)

- Developed data-processing pipelines for a reimplementation of Google DeepMind's MetNet-3 to support efficient training on large-scale spatiotemporal data
- Benchmarked distributed training configurations to characterize trade-offs between execution time and memory usage

Jun. 2024 – Jun. 2025 — **Preserving Feature Dependencies in Synthetic Tabular Data**
Undergraduate Researcher | AI Education Institute, Hansung University

- Developed BN-CTGAN by integrating Bayesian networks with CTGAN to preserve feature dependencies in synthetic tabular data, reducing mean PMI error by 91.96% relative to CTGAN

Mar. 2024 – Oct. 2024 — **Load-Aware Multipath Forwarding in Distributed Networks**
Undergraduate Research Assistant | Information Systems Lab, HUFS

- Designed and implemented PFS-DFF, applying proportional-fair scheduling to distribute data fragments across paths based on link throughput and load conditions
- Evaluated PFS-DFF on a distributed Raspberry Pi testbed, reducing end-to-end data transfer time by ~9% versus a single-path baseline under high network load

Sep. 2022 – May 2023 — **LSTM-Based Traffic Forecasting in Distributed Networks**
Undergraduate Research Assistant | Mobile Distributed Computing Lab, HUFS

- Developed per-node LSTM models to forecast traffic demand in a distributed Raspberry Pi network, achieving test-set R^2 of 0.94–0.96 across three nodes

PUBLICATIONS

[1] E. Kim, **K. Lee**, Y. Kim, H. Jung, H. Choi, I. Kim, and H. Park. Frequency-Aware Multi-Resolution Architecture for High-Resolution Spatiotemporal Map Reconstruction from Sparse Observations. *38th Workshop on Image Processing and Image Understanding (IPIU 2026)*, 2026, Jeju, Korea (Poster)

[2] Y. Kim, E. Kim, **K. Lee**, H. Park, H. Choi, I. Kim, and H. Jung. Learnable Adjacency Matrix Based Spatio-Temporal Graph Convolutional Network for Traffic Prediction. *Autumn Annual Conference of the Institute of Electronics and Information Engineers (IEIE 2025)*, 2025, Gyeonggi, Korea (Poster)

[3] **K. Lee**, J. Baik, H. Ryou, S. Oh, and I. Doo. A Comparative Study of Synthetic Data with Generative AI Models: Safely Reproducing Sensitive Transportation Worker Data. *National Conference on Advanced Engineering and ICT-Convergence (NCAEIC 2023)*, 2023, Jeju, Korea (Oral, **Best Paper Award**)

PRESENTATION & DEMONSTRATION

Oct. 2025 — **Korea Police World Expo (KPEX) 2025** — Incheon, Republic of Korea
Research Presenter, representing Visual Intelligence Group, KIST
- Presented the objectives, methodology, and public-safety impact of our real-time crowd density prediction system to ~200 booth visitors

TEACHING & MENTORSHIP

Mar. 2024 – Jun. 2024 — **Teaching Assistant, Network Security** | HUFS
Supervisor: Prof. Seong Ho Jeong
- Graded assignments and provided feedback for 30+ students through Q&A sessions

Sep. 2022 – Dec. 2022 — **Programming Mentor, AdvICE Programming Club** | HUFS
Mentee: Sangyeon Park (Undergraduate at HUFS)
- Mentored a student in Python programming, algorithm design, and competitive programming

Sep. 2021 – Dec. 2021 — **Teaching Assistant, Algorithm Analysis and Design** | HUFS
Supervisor: Dr. Sayed Chhattan Shah
- Provided detailed code assignment reviews and debugging support for 50+ students

SCHOLARSHIPS

Mar. 2024	President's Scholarship (Full tuition), HUFS
Sep. 2023	Hyoryeong Merit Scholarship, Cheong-gwon-sa Scholarship Foundation
Sep. 2023	Academic Merit Scholarship (One-quarter tuition), HUFS
Mar. 2023	Dean's Scholarship (Half tuition), HUFS
Sep. 2022	President's Scholarship (Full tuition), HUFS

AWARDS

Dec. 2023	Best Paper Award, NCAEIC 2023
Dec. 2023	Excellence Award, Capstone Design Project, HUFS
Dec. 2023	Excellence Award, Data Creator Camp, Ministry of Science & ICT
Sep. 2020	Creative Award, National Defense Start-Up Challenge, Top 25 of 939 teams

TECHNICAL SKILLS

Programming & Systems	Python, C/C++, Bash, Linux, Git
Networking & Distributed Systems	PyTorch Distributed, DeepSpeed, Wireshark, ns-3, hping3
Machine Learning & Data	PyTorch, TensorFlow, Keras, NumPy, Pandas, Matplotlib
Experimental Infrastructure	CloudLab, ACCESS, Raspberry Pi